Klayver Paz

AI Engineer · LLM Agents & RAG

klayverpaz@gmail.com +55 85 99692-9159 Fortaleza, Brazil · Remote
linkedin.com/in/klayverpaz github.com/klayverpaz Portfolio

AI Engineer building production LLM agents and RAG systems: LangGraph multi-agent orchestration, retrieval over PostgreSQL + pgvector, evaluation harnesses, and the FastAPI / React / AWS services around them. Founder and sole engineer of Seegold, a SaaS live with 22 paying gyms. Research roots in computer vision and geospatial ML at NYU. Based in Fortaleza, Brazil; remote with US time-zone overlap.

EXPERIENCE

AI Engineer / Agent Engineer *Arke Analytica* Jan 2026 – Present

Production LLM agents and RAG systems for B2B platforms. Independent practice.

- **Multi-agent system (LangGraph).** Reduced a recurring B2B operational workflow from ∼1 month to ∼40 minutes with a production multi-agent system on LangGraph: agents only where the work is open-ended (a ReAct planner with 14 tools), deterministic workflow subgraphs everywhere else, and human-in-the-loop approval gates before any state-changing operation.
- **RAG pipeline.** Designed three-tier RAG retrieval over the client's own history (current project → same company → general knowledge base on PostgreSQL + pgvector), with deterministic metadata filters applied before cosine similarity on OpenAI embeddings, so the agent only reasons over relevant context.
- **LLM evaluation.** Built an accuracy harness for LLM pipelines: real cases as fixtures, hit-rate / leakage / miss metrics, repeated runs per case compared against a blessed baseline per model and commit, plus LangSmith tracing and golden conversation tests on the production model.
- **Agent safety & human-in-the-loop.** Interrupt-and-resume pattern lets users pause an in-flight workflow, ask a side question, and resume from the same node, backed by TTL-bounded Redis checkpoints. Bearer-token pass-through: the agent never stores credentials and every tool acts with the caller's own permissions.
- **Shipped on AWS.** FastAPI + async SQLAlchemy + PostgreSQL services (Anthropic / OpenAI provider abstraction, SSE streaming) and a React 19 + TypeScript frontend, containerized to Amazon ECR via Bitbucket Pipelines and deployed GitOps-style to Kubernetes with Argo CD (dev / prod), using S3 and TLS-secured Redis.

Stack: LangGraph, LangChain, LangSmith, Anthropic Claude, OpenAI, RAG, pgvector, PostgreSQL, Redis, FastAPI, React 19, TypeScript, AWS (ECR, S3), Kubernetes, Argo CD.

Research Scientist (Data Science) *Vida Center — NYU Tandon School of Engineering* Sep 2025 – Dec 2025

Contributed to the UrbanMapper project — analytics framework for large-scale urban Big Data processing and spatial-temporal modeling.

- **CRIF library.** Built CRIF (Calibrated Raster Interpolation), a Python library for unified geospatial rasterization and dasymetric mapping — formal mathematical derivation, custom Jupyter widgets, Docker packaging.
- **Bayesian + geospatial.** Applied Bayesian modeling and geospatial methods (GeoPandas, Rasterio) to urban Big Data: point-to-raster aggregation, polygon rasterization with overlap handling, hotspot discovery via raster fusion.
- **Cross-discipline work.** Collaborated with urban science, data visualization, and applied math teams to enhance data-driven decision-making in research contexts.

Stack: Python, GeoPandas, Rasterio, PyTorch, Bayesian modeling, Jupyter widgets, Docker.

Data Scientist *Atlântico* Nov 2024 – Aug 2025

LLM-powered NLP and anomaly-detection systems for industrial datasets.

- **LLM / NLP pipelines.** Shipped sentiment analysis, text classification and entity recognition over large-scale industrial text, built with LangChain and LangGraph on OpenAI GPT models, using RAG for domain context and LangSmith for tracing and evaluation.
- **Anomaly detection.** Built defect-forecasting models for industrial time-series data.
- **APIs & microservices.** Delivered Python and .NET services on Azure.

Stack: Python, LangChain, LangGraph, LangSmith, OpenAI, RAG, Azure, .NET.

Researcher (Computer Vision) *Federal University of Ceará (UFC)* Sep 2023 – Aug 2025

Computer vision research applied to highway infrastructure management, within Ceará's Chief Scientist Program (FUNCAP).

- **MIDR ensemble.** Built a YOLO-ensemble defect-detection system with custom IOU/NMS fusion logic. Patented and currently the standard inspection system used by the Court of Accounts of the State of Ceará (TCE-CE).
- **Data pipelines.** Designed scalable storage, preprocessing, and modeling pipelines for highway imagery datasets.
- **Iterative refinement.** Collaborated with civil engineering and AI experts to refine defect-detection accuracy across model generations.

Stack: PyTorch, YOLO, Ultralytics, OpenCV, Python.

Data Scientist & Software Developer *Agilean* Oct 2020 – Nov 2024

Predictive analytics and data-integration systems for the construction and industrial sectors.

- **Predictive ML.** Forecasted construction-site waste and schedule delay using ML and data-mining over operational data.
- **ETL pipelines.** Designed and deployed integrations on Azure Data Factory.
- **APIs & microservices.** Built C# ASP.NET MVC and Python services on Azure Cloud.

Stack: Python, ML, Azure, ETL, C# / .NET.

FEATURED PROJECTS

Multi-Agent Operations Assistant *Production — Confidential (under NDA)*

LangGraph multi-agent system that cut a recurring B2B operational workflow from ∼1 month to ∼40 minutes (∼99%). Intent classification, deterministic workflow subgraphs, HITL gates, lateral-query interrupt-and-resume, Redis checkpoints, deployed on AWS + Kubernetes. *LangGraph, LangSmith, pgvector, Redis, FastAPI, React 19, AWS (ECR, S3), Kubernetes + Argo CD.*

Seegold — SaaS + WhatsApp AI Agent *Production — 22 paying gyms*

Multi-tenant billing SaaS for gyms and martial-arts schools (automatic charges and payment receipts over WhatsApp), built and operated end to end as founder and sole engineer. Includes an LLM sales agent that drafts WhatsApp replies to leads: deterministic guardrails (whitelisted prices and links, human review for anything else), prompt-injection containment (model runs with no tools), and send verification against the outbound record. *FastAPI, SQLAlchemy async, PostgreSQL, React 19, Anthropic Claude, Twilio WhatsApp, Asaas, AWS (ECS, RDS, CloudFront), Terraform.*

Personal Finance Skill *In development — Anthropic Skill*

Anthropic Skill that processes Brazilian bank statement PDFs through a stage-based state machine: parse → normalize → validate → classify → persist. Idempotent imports, self-improving merchant classifier. *Anthropic Skills protocol, Python, SQLite, PDF parsing.*

CRIF — Calibrated Raster Interpolation *Open research*

Geospatial Python library for unified grid rasterization, dasymetric mapping, and hotspot discovery. Point + polygon rasterization with overlap handling and multi-band normalization, formal mathematical derivation, custom Jupyter widgets. Built at NYU VIDA Center for the UrbanMapper project. *Python, GeoPandas, Rasterio, NumPy, Jupyter widgets, Docker.*

MIDR — Pavement Defect Detection *Production — Adopted*

YOLO-ensemble defect detection deployed in highway inspection by the Court of Accounts of Ceará (TCE-CE). Custom IOU + NMS fusion across multiple class-specialized YOLO models (potholes, cracks, patches, signage); per-version SQLite-backed dataset versioning. *Python, PyTorch, Ultralytics YOLO, OpenCV.*

Nutrir — Meal Verification CV *Production — Adopted*

Image-segmentation pipeline that classifies meal components (rice, beans, meat, vegetables) from a photo and pairs the percentages with a calibration-scale weight to produce structured per-meal nutritional records. Adopted by the Municipality of Fortaleza. *Python, PyTorch, image segmentation, OpenCV.*

SKILLS

Core (Agent / LLM Engineering): LangGraph, LangSmith, LangChain, multi-agent orchestration, RAG, pgvector, HITL workflows, LLM evaluation, prompt-injection guardrails, OpenAI / Anthropic SDKs, Amazon Bedrock (Converse API, Knowledge Bases), MCP, Anthropic Skills protocol, Claude Code (agentic coding)

Backend Engineering: Python 3.12, FastAPI, Pydantic v2, SQLAlchemy 2.0 async, Alembic, PostgreSQL / SQL Server, Redis, DDD + CQRS, Value Objects, vertical slicing

Frontend Engineering: TypeScript, React 19, Vite, Tailwind v4, TanStack Query v5, React Router v7, Radix UI, react-hook-form + zod, Vitest, MSW

DevOps & Quality: Docker, CI/CD (Bitbucket Pipelines, GitHub Actions), pytest, ruff, mypy, type-checked Python, AWS (ECS, RDS, ECR, S3, CloudFront, CloudWatch), Terraform, Kubernetes + Argo CD

Computer Vision & Geospatial ML: PyTorch, YOLO (ensemble + custom fusion), OpenCV, scikit-learn, Bayesian modeling, GeoPandas, Rasterio, pandas / NumPy

Past stack (still serviceable): C# / .NET, Azure, Power BI, Microsoft Fabric, Databricks

EDUCATION

B.S. Civil Engineering (Computer Science research track) *Universidade Federal do Ceará (UFC)* 2022
Concurrent research training in Data Science via the Chief Scientist Program (FUNCAP, Ceará State Government).
Selected coursework (CS track): Machine Learning, Data Mining, Digital Image Processing, Numerical Methods, Programming Fundamentals, Linear Algebra, Probability & Statistics, Optimization.

Research Scientist (Data Science) *NYU Tandon School of Engineering — Vida Center* Sep 2025 – Dec 2025
Urban analytics and geospatial ML research affiliation (UrbanMapper / CRIF).

LANGUAGES

English — Full professional proficiency | **Portuguese** — Native speaker